

B.Tech (Computer Science and Engineering) Semester-VII

Subject: Data Mining & Data Warehouse

Marks: 64

ASSIGNMENT #3

Section A (2 Marks Each)

[CO3] Q1. What is an association rule?

Answer:

An association rule is a data mining rule that shows a relationship between items in a dataset, usually in the form $X \rightarrow Y$, meaning if X occurs, Y is likely to occur. It is used to identify patterns and co-occurrences in large databases, especially in market basket analysis.

[CO3] Q2. Define frequent pattern.

Answer:

A frequent pattern is a pattern (itemset, subsequence, or substructure) that appears in a dataset frequently, i.e., its support value is greater than or equal to a minimum support threshold defined by the user.

[CO3] Q3. What does iterative approach mean in Apriori algorithm?

Answer:

The iterative approach in the Apriori algorithm refers to its **level-wise search**, where it starts by finding frequent 1-itemsets, then uses them to generate 2-itemsets, then 3-itemsets, and so on until no more frequent patterns exist. Each iteration uses the results of the previous step.

[CO3] Q4. Write applications of association rules.

Answer:

Applications of association rules include:

1. Market Basket Analysis
2. Recommender Systems
3. Fraud Detection

4. Web Usage Mining
 5. Medical Diagnosis
 6. Customer Behavior Analysis
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[CO3] Q5. Label the association rule $X \rightarrow Y$ [Support 50%, Confidence 70%].

Answer:

- **Support (50%):** 50% of the total transactions contain both X and Y.
 - **Confidence (70%):** 70% of the transactions that contain X also contain Y.
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Section B (7 Marks Each)

[CO3] Q1. Define support and confidence in relation to frequent pattern.

Answer:

- **Support:**
Support of an itemset is the percentage or proportion of transactions in the database that contain the itemset.

$$\text{Support}(X) = \frac{\text{No. of transactions containing } X}{\text{Total transactions}}$$

- **Confidence:**
Confidence of a rule $X \rightarrow Y$ is the likelihood that Y appears in a transaction that contains X.

$$\text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)}$$

Support tells frequency, confidence tells strength of implication.

[CO3] Q2. Briefly describe correlation analysis.

Answer:

Correlation analysis in data mining measures how strongly two items or events are related, beyond support and confidence. It determines whether the presence of one item increases, decreases, or has no effect on the presence of another.

- It uses statistical measures like **Lift, Chi-square, Cosine, and Conviction**.

- If two items appear together frequently but are not statistically dependent, the rule may be misleading.
 - High correlation means a strong true relationship, not just a coincidence.
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[CO3] Q3. Explain market basket analysis and its uses.

Answer:

Market Basket Analysis (MBA) is a data mining technique used to analyze customer purchase behavior by finding associations between products purchased together.

Uses:

1. **Product Placement:** Grouping related products in stores.
 2. **Cross-selling & Recommendations:** e.g., Amazon's "*Customers who bought this also bought...*".
 3. **Bundling & Promotions:** Offering discounts on frequently bought product pairs.
 4. **Inventory Management:** Stocking products that are purchased together.
 5. **Customer Behavior Insights:** Understanding buying patterns to improve sales strategies.
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Section C (11 Marks Each)

[CO3] Q1. Explain association rules with the help of Apriori algorithm using suitable examples.

Answer:

Association rule mining is a technique used to discover interesting relations between variables in large databases. The **Apriori algorithm** is one of the earliest and most popular algorithms for mining frequent itemsets and generating association rules.

✅ Steps in Apriori Algorithm

1. **Set minimum support & confidence**
2. **Generate all 1-itemsets** and calculate support
3. **Select only those itemsets whose support $\geq$ minimum support** (Frequent 1-itemsets)
4. **Generate 2-itemsets** from frequent 1-itemsets
5. **Prune itemsets that contain infrequent subsets (Apriori property)**

6. **Repeat for k-itemsets** until no more frequent itemsets are found
 7. **Generate association rules** from frequent itemsets
 8. **Calculate confidence of each rule and keep those $\geq$ minimum confidence**
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✅ **Example Dataset**

Transaction ID	Items Purchased
T1	Milk, Bread
T2	Milk, Diaper, Beer, Bread
T3	Bread, Butter
T4	Milk, Bread, Butter
T5	Bread, Diaper, Milk, Beer

Assume **Minimum Support = 40% (2 transactions)**, **Minimum Confidence = 60%**

✅ **Step-by-Step Execution**

Step 1: Generate 1-itemsets with support

Item	Support Count
Milk	4
Bread	5
Butter	2
Diaper	2
Beer	2

✅ All have support $\geq 2 \rightarrow$ All are **Frequent 1-itemsets (L1)**

Step 2: Generate 2-itemsets (C2) and calculate support

Itemset	Support Count
Milk, Bread	3

Itemset	Support Count
Milk, Diaper	2
Milk, Beer	2
Bread, Butter	2
Bread, Diaper	2
Diaper, Beer	2

✓ All have support $\geq 2 \rightarrow$ **Frequent 2-itemsets (L2)**

Step 3: Generate 3-itemsets (C3)

Itemset	Support Count
Milk, Bread, Diaper	2
Milk, Diaper, Beer	2
Bread, Diaper, Beer	1 ✗ (remove)

✓ Only 2 itemsets are frequent $\rightarrow$ **Frequent 3-itemsets (L3)**

Step 4: Generate Association Rules

Example Rule:

Rule: Milk $\rightarrow$ Bread

Support = $3/5 = 60\%$

Confidence = $\text{Support}(\text{Milk} \cup \text{Bread}) / \text{Support}(\text{Milk}) = 3/4 = 75\%$ ✓ ($\geq 60\%$)

✓ Accepted

Another Rule:

Rule: Diaper $\rightarrow$ Beer

Support = $2/5 = 40\%$

Confidence = $2/2 = 100\%$ ✓

✓ Strong rule

✓ **Final Strong Association Rules**

Rule	Support	Confidence
Milk → Bread	60%	75%
Diaper → Beer	40%	100%
Bread → Milk	60%	60%
Milk, Diaper → Beer	40%	100%

✓ Advantages of Apriori

- ✓ Simple and easy to implement
- ✓ Uses downward closure property for pruning
- ✓ Works well on small to medium datasets

✓ Limitations

- ✗ Generates huge number of candidate sets
- ✗ Multiple database scans increase time complexity
- ✗ Slow for large datasets → FP-Growth solves this

[CO3] Q2. Briefly explain FP-Growth algorithm using suitable example.

Answer:

FP-Growth (**Frequent Pattern Growth**) is an efficient algorithm for mining frequent patterns without generating candidate itemsets like Apriori. It compresses the database into a **compact FP-tree** and extracts frequent itemsets using **recursive tree mining**.

✓ Steps in FP-Growth

1. Scan database and find support of each item
2. Remove infrequent items
3. Sort items in descending support order
4. Build FP-tree (prefix tree)
5. Extract conditional pattern base for each item
6. Recursively generate frequent itemsets

✓ Example Dataset

(Same as Apriori example)

TID	Items
T1	Milk, Bread
T2	Milk, Diaper, Beer, Bread
T3	Bread, Butter
T4	Milk, Bread, Butter
T5	Bread, Diaper, Milk, Beer

✅ Step 1: Count item frequency

Item	Frequency
Bread	5
Milk	4
Diaper	2
Beer	2
Butter	2

✅ Step 2: Build FP-Tree

Tree nodes arranged in descending order of frequency:

(Root)
/
Bread(5)
/ \
Milk(3) Butter(2)
|
Diaper(2)
|
Beer(2)

✓ FP-Growth Mining

- Conditional pattern base for **Beer**: {Milk, Diaper}
- Frequent patterns: (Beer, Diaper), (Beer, Milk, Diaper)

✓ Advantages of FP-Growth

- ✓ No candidate generation
- ✓ Uses compressed tree → less memory
- ✓ Faster than Apriori for large datasets

✓ Disadvantages

- ✗ FP-tree may still be large
- ✗ Hard to implement compared to Apriori
- ✗ Tree structure depends on item ordering

[CO3] Q3. Define Normalization. Also discuss (a) Min-Max normalization (b) Z-score normalization.

Answer:

✓ Definition of Normalization

Normalization is a data preprocessing technique used to scale numerical values into a standard range so that no variable dominates the model due to its magnitude. It is essential in machine learning, clustering, and distance-based algorithms.

✓ (a) Min-Max Normalization

It rescales data into a fixed range, usually **[0,1]** or **[-1,1]**.

Formula:

$$v' = \frac{v - \min}{\max - \min}$$

Example:

Given values = [10, 20, 30, 40], normalize 20 in range [0,1]:

$$v' = \frac{20 - 10}{40 - 10} = \frac{10}{30} = 0.33$$

✓ Preserves the original distribution but compresses values into a range.

✓ (b) Z-Score Normalization (Standardization)

It transforms data based on mean (μ) and standard deviation (σ). Output mean = 0, SD = 1.

Formula:

$$v' = \frac{v - \mu}{\sigma}$$

Example:

Dataset values: [10, 20, 30], mean = 20, SD = 8.16

Normalize 30:

$$v' = \frac{30 - 20}{8.16} = 1.22$$

✓ Used when data has outliers

✓ Useful in clustering, PCA, regression

✓ Difference between Min-Max and Z-Score Normalization

Feature	Min-Max	Z-Score
Range	[0,1]	No fixed range
Sensitive to Outliers	Yes	No
Preserves Distribution	Yes	No
Use Case	Neural nets, image scaling	ML algorithms like SVM, KNN